Fables-DRT

Guidelines for the Annotation of FABLES-DRT
Version 1.0

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Preface

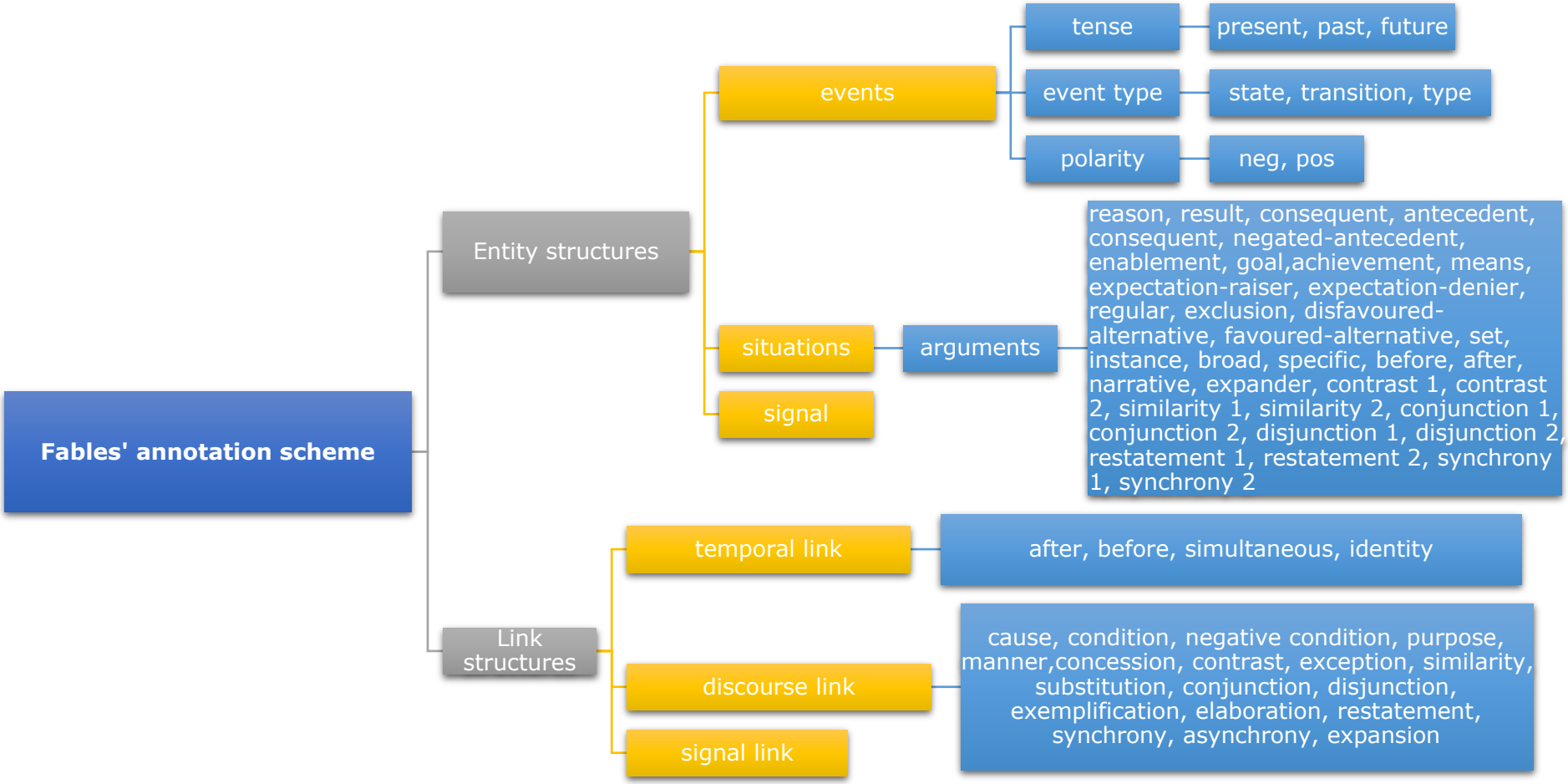
This manual describes the process of a semantic annotation scheme designed for analyzing temporal and rhetorical information in a corpus of fables in Portuguese, *Fables-DTR: A Corpus of Fables Annotated for Discourse and Temporal Relations*. It was built based on the Text2Story manual of annotation (Silvano et al. 2023; Nunes et al. 2024). The proposed scheme is based on the Text2Story annotation scheme (Silvano et al. 2021; Leal et al. 2022, Silvano et al. 2025), MDAI annotation scheme (Tomaszewska et al. 2024). This scheme harmonizes parts 1 and 8 of ISO 24617 *Language resource management - Semantic annotation framework.* This annotation framework includes entity structures (events, situations, signals) and links (temporal and discourse) with tags and attribute values.

Sources

ISO24617-1:2012, Language resource management-Semantic annotation framework (SemAF) - Part 1: Time and events (SemAF-Time, ISO-TimeML)

ISO-24617-8: 2016, Language resource management- Semantic annotation framework (SemAF) - Part 8: Semantic relations in discourse (SemAF-DR-Core)

I. Semantic annotation framework



II. Stages of the annotation process

1st stage	Annotation of the discourse relations arguments with the label situations
	Annotation of the signal
	Annotation of the discourse relations arguments with the semantic role
	Annotation of the discourse link
	Annotation of the signal link
2nd stage	Annotation of the main event of each argument
	Annotation of the events attributes
	Annotation of the temporal link

III. Annotation of the discourse relation

A. Entity structures

A.1. Situations

Definition

- Eventualities represented by the discourse relations arguments.
  Ex. **John stayed home** because he was sick.
  This sentence has two situations marked in bold and underlined.

Tag span (markable)

- The complete argument of a discourse relation

Notes:

1. The signal, whenever present, is not included in the markable of the situation.

2. To ensure that no part of the speech is left unnoted, the direct speech must be inserted within the markable, including the reporting verb (e.g., *say, ask, answer*) that introduces that same direct speech (typically marked with quotation marks).
   Ex. The Ants stopped work for a moment, though this was against their principles.
   SIT1["May we ask," said they, "what you were doing with yourself all last summer? Why didn't you collect a store of food for the winter?"]
   SIT2["The fact is," replied the Grasshopper, "I was so busy singing that I hadn't the time."]
   SIT3["If you spent the summer singing," replied the Ants, "you can't do better than spend the winter dancing."]

3. The identification of the markable of the situations must exclude sentence punctuation. In the following example, SIT1 does not include the comma, and SIT2 does not include both the comma and the period.
   Ex.
   SIT1[One fine day in winter some Ants were busy drying their store of corn]**,** which SIT2[had got rather damp during a long spell of rain]**.**

The rule of excluding punctuation at the beginning or end of a situation's markable does not apply to quotation marks when they are used to delimit direct speech. In this case, the quotation marks must be included in the situation's markable (as well as all punctuation in the discourse segment delimited by the quotation marks).

Ex.… Presently up came a Grasshopper and begged them to spare her a few grains, SIT**[“For,” she said, “I’m simply starving.”]** The Ants stopped…

4. When there is adverbial subordination in the text, not only must each of the clauses (subordinate and adverbial subordinate) be annotated, but also the unit formed by the combination of the two clauses.

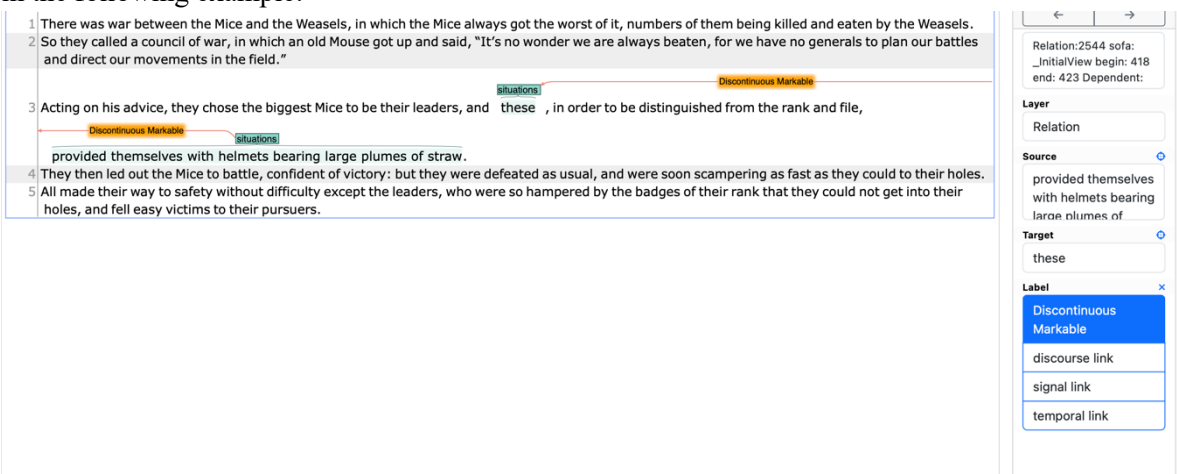
Ex. A thirsty Crow found a Pitcher with some water in it, but so little was there that, **try as she might, she could not reach it with her beak**, and it seemed as though she would die of thirst within sight of the remedy.

SIT[try as she might]

SIT[she could not reach it with her beak]

SIT[try as she might, she could not reach it with her beak]

5. When the markable of the situation of discontinuous, the annotator should annotate the two parts and link them with the technical link “discontinuous markable”, as illustrated in the following example:



6. The annotator should identify as many discourse relations as necessary, and the same situation can be an argument for more than one discourse relation. In other words, the same unit (situation) can have different roles depending on the DR it's involved in. In those cases, we may need to annotate twice the same markable of an argument or part of it.

Ex. *A wolf decided to disguise itself so it could attack a flock of sheep without fear of discovery. So, it dressed in sheep's clothing and slipped among the sheep while they were grazing.*

<u>Cause</u>

Reason: to be able to attack a flock of sheep without fear of discovery

Result: it dressed in sheep's clothing

<u>Purpose</u>

Enablement: A wolf decided to disguise itself

Goal: to be able to attack a flock of sheep without fear of discovery

Asynchrony
before: it dressed in sheep's clothing
after: it slipped among the sheep while they were grazing

Attributes

- Arguments semantic role: identified in Table 1.

A.2. Signals

Definition

- Connectors that link Arg 1 to Arg 2 and that mark explicitly the discourse relation arguments. Ex. **John stayed home** because he was sick.

Tag span (markable)

- A word or expression that links two arguments and signals the discourse relation. (e.g. *because, due to, which, and, or,…*).

	DR-core relations	Definition	Semantic Role	
			Arg 1	Arg2
asymmetric	**Cause**	Arg2 is an explanation for Arg1.	result	reason
	Expansion	Arg2 is a situation involving some entity/entities in Arg1, expanding the narrative of which Arg1 is a part, or expanding on the setting relevant for interpreting Arg1. The Arg1 and Arg2 situations are distinct.	narrative	expander
	Asynchrony	Arg1 temporally precedes Arg2.	before	after
	Concession	An expected causal relation between Arg1 and ¬Arg2 is cancelled or denied by Arg2.	expectation raiser	expectation-denier
	Elaboration	Arg1 and Arg2 are the same situation, but Arg2 provides more detail.	broad	specific
	Exemplification	Arg1 is a set of situations; Arg2 is an element of that set.	set	instance
	Manner	Arg2 specifies how Arg1 comes about or occurs.	achievement	means
	Condition	Arg2 is an unrealized situation which, when realized, would lead to Arg1.	Consequent	Antecedent
	Negative Condition	Arg2 is an unrealized situation which, when "not" realized, would lead to Arg1.	Consequent	Negated-Antecedent
	Purpose	Arg2 is the goal or purpose of the situation described by Arg1.	Enablement	Goal
	Exception	Arg2 indicates one or more circumstances in which the situation(s) described by Arg1 does not hold.	Regular	Exclusion
	Substitution	Arg1 and Arg2 are alternatives, with Arg2 being the favoured or chosen alternative.	Disfavoured-alternative	Favoured-alternative

	DR-core relations	Definition	Semantic roles	
			Arg 1	Arg2
symmetric	**Conjunction**	Arg1 and Arg2 bear the same relation to some situation evoked in the discourse, explicitly or implicitly. Their conjunction indicates that they both hold with respect to that situation.	Conjunction 1	Conjunction 2
	Contrast	One or more differences between Arg1 and Arg2 are highlighted with respect to what each predicate as a whole or to some entities they mention.	Contrast 1	Contrast 2
	Synchrony	Some degree of temporal overlap exists between Arg1 and Arg2. All forms of overlap are included.	Synchrony 1	Synchrony 2
	Similarity	One or more similarities between Arg1 and Arg2 are highlighted with respect to what each predicate as a whole or to some entities they mention.	Similarity 1	Similarity 2
	Disjunction	Arg1 and Arg2 bear the same relation to some other situation evoked in the discourse, explicitly or implicitly. Their disjunction indicates that they are alternatives with respect to that situation, with the disjunction being non-exclusive so that both Arg1 and Arg2 may hold.	Disjunction 1	Disjunction 2
	Restatement	Arg1 and Arg2 describe the same situation, but from different perspectives.	Restatement 1	Restatement 2

Table 1. List of discourse relations and semantic roles

B. Link structure

B.1. Discourse Link

Definition

- The semantic connection that is established between two arguments. The list of discourse relations and their definition is presented in Table 1.

Notes:

1. The link between markables (whether it is the annotation of situations or events) must always be made from the second markable to the first markable in the linear order of the discourse.

   Ex.

   Expansion

   SIT1[One fine day in winter some Ants were busy drying their store of corn], which SIT2[had got rather damp during a long spell of rain].

   SIT1 ← (Expansion) SIT2

2. The annotator should only establish a discourse link between two situations linked by one of the discourse relations presented in Table 1. This implies that, whenever there are that-clauses, comparative clauses, the two situations are left without any annotation.
   a. Ex. *he ran as fast as he could.*
      i. *He said that he was afraid.*

B.2. Signal Link

Definition

- It's a link that establishes the connection between the signal and Arg of the discourse relation that is introduced by that signal.

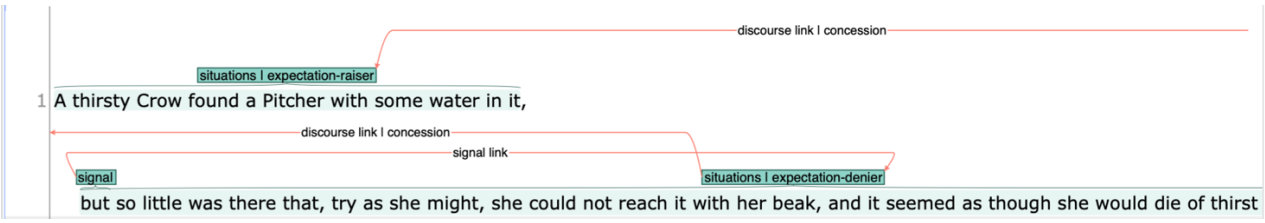


Figure 1. Example of phase 1 of annotation- Discourse relation layer.

III. Annotation of the temporal relation (Silvano et al. 2023)

C. Entity structures

C.1. Events

Definition

- eventuality that happens or occurs or state or circumstance that is temporally relevant, i.e., that is directly related to a temporal expression or change over the course of the text.

Tag span

- verb forms (only main verbs, not auxiliary verbs). Ex. *The police arrested the burglar. John has visited his aunt.*

Notes:

- in case of a predicative construction, we leave the copular verb and mark the adjective/ noun/ prepositional complement, identifying the POS as an adjective or noun, but assigning the values of the tags TENSE according to the information from the verb. Ex. *The police were ready.*
  If the copula verb is followed by a prepositional or adverbial phrase with an interpretation of spatial or temporal location, mark only the verb. Ex. *John was there. John is in Lisbon*.
- if the verb is included in a (semi) lexicalized expression, mark that expression as the predicator. Ex. *The suspect will face charges.*

Attributes

- type:
  - STATE – a situation in which something obtains or holds true. Ex. *John was sick.*
  - PROCESS – a situation that is dynamic and atelic, with duration. Ex. *John swam.*
  - TRANSITION – a situation that is dynamic, with duration and consequent state. It subsumes accomplishments and achievements. Ex. *John read a book/ John won the race.*

- tense:
  - PRESENT – locates a situation at the present time, in the near future or in a time interval that starts in the past and still holds in the present. It can also depict habitual situations.
  - PAST – locates situations in a past time.
  - FUTURE – locates situations in the future.

- polarity:
  - NEG
  - POS

D. Link structure

D.1. Temporal Link

Definition

- establishes different temporal relationships (simultaneous, after, before, identity) between two events
- Directionally: the link between markables (whether it is the annotation of situations or events) must always be made from the second markable to the first markable in the linear order of the discourse.

Ex.

After

… Presently up EVENT1[came] a Grasshopper and EVENT2[begged] them to spare her a few grains…

EVENT1 ← (After) EVENT2

Attributes

- BEFORE – a situation/ time expression occurs before another one.
- AFTER – a situation/ time expression occurs after another one.
- SIMULTANEOUS – a situation/ time expression occurs at the same time as another one.
- IDENTITY – two situation are the same.

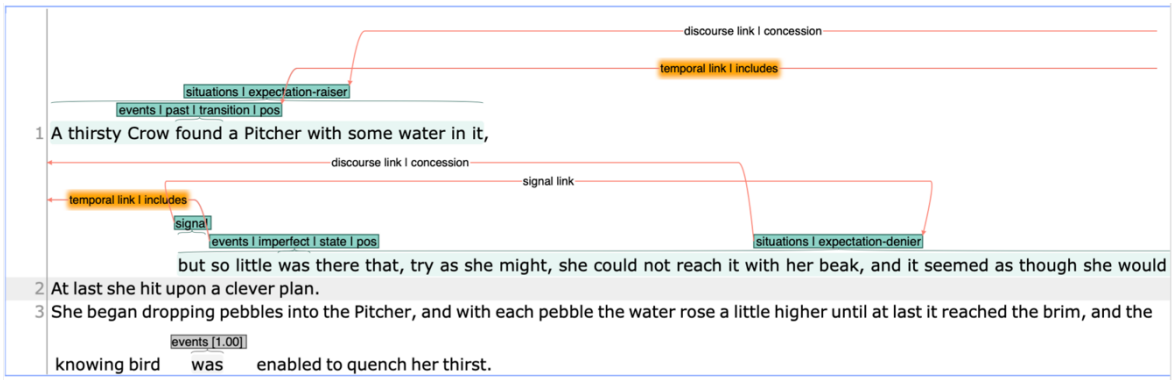


Figure 2. Example of phase 2 of annotation- Temporal relation layer

Publications using the annotation framework

Silvano, Purificação; Leal, António; Silva, Fátima; Cantante, Inês; Oliveira, Fátima & Jorge, Alípio (2021). Developing a multilayer semantic annotation scheme based on ISO standards for the visualization of a newswire corpus. In: *Proceedings of The Seventeenth Joint ACL - ISO Workshop on Interoperable Semantic Annotation*. Groningen, Netherlands.

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Tomaszewska, Aleksandra, Silvano, Purificação, Leal, António and Amorim, Evelin. 2024. ISO 24617-8 Applied: Insights from Multilingual Discourse Relations Annotation in English, Polish, and Portuguese. In *Proceedings of the 20th Joint ACL - ISO Workshop on Interoperable Semantic Annotation @ LREC-COLING 2024*, pages 99–110, Torino, Italia. ELRA and ICCL.